

MATH1231 CALCULUS

Chapter 5

Averages, arc length, speed, and surface area

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A/Prof Mark Holzer

UNSW Sydney
mholzer@unsw.edu.au

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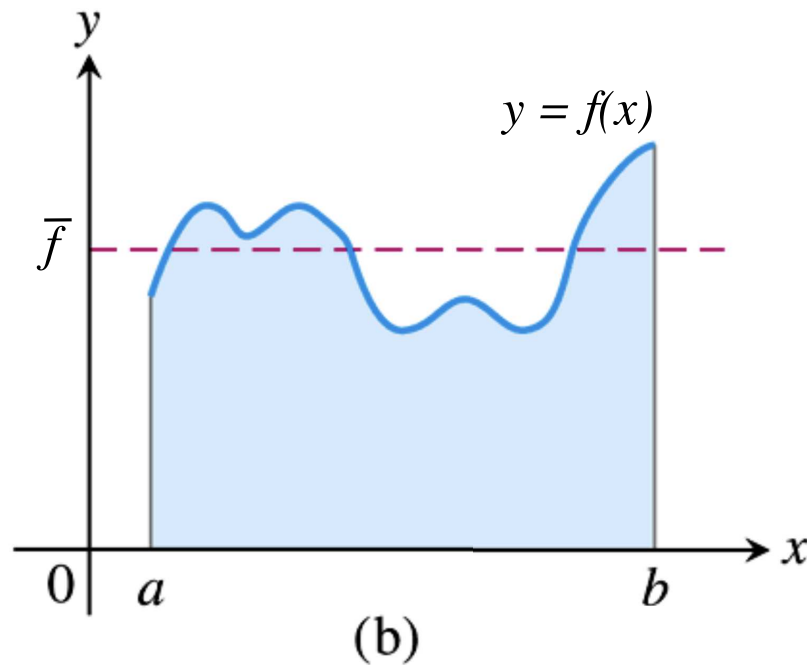
Overview

- ▶ How do we find the average value of a function?
- ▶ How do we calculate the length of a curve?
- ▶ How do we calculate the velocity and speed of a particle travelling along a curve?
- ▶ How do we calculate area and volume of a solid outlined by spinning the curve $y = f(x)$ around the x (or y) axis?

Average or mean value of a function

If $f(x)$ is integrable on $[a, b]$, then the *average value* (also called the *mean value*) of $f(x)$ over the interval $x \in [a, b]$ is given by

$$\bar{f} = \frac{1}{b-a} \int_a^b f(x) dx .$$



Example

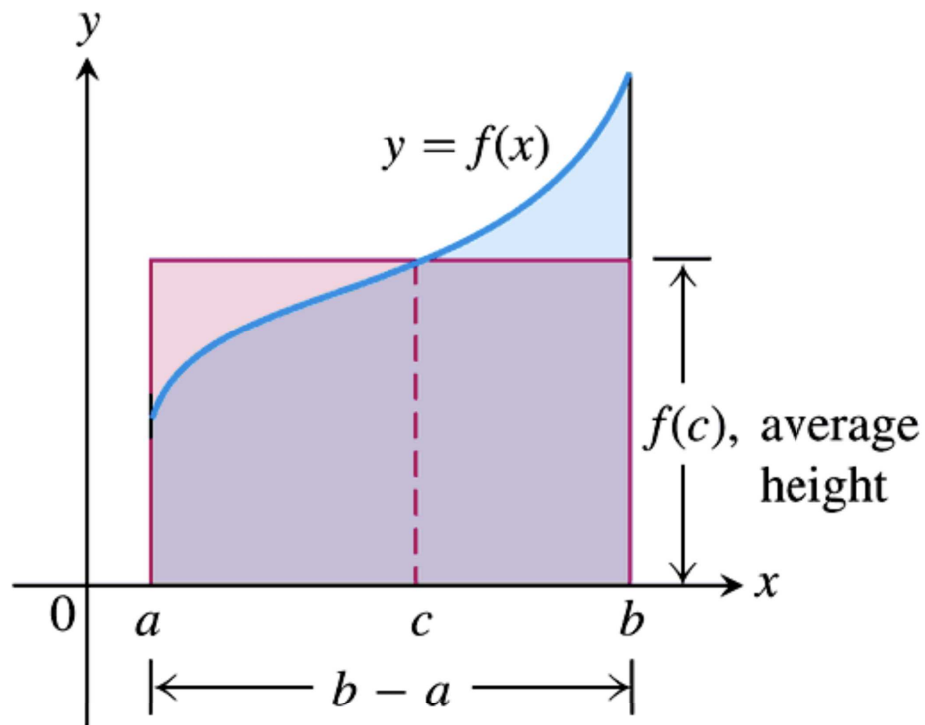
Find the average value of $f(x) = \sqrt{4 - x^2}$.

Mean value theorem for integrals

Theorem: If $f(x)$ is continuous on $[a, b]$, then at some point $c \in [a, b]$

$$f(c) = \frac{1}{b-a} \int_a^b f(x) dx \quad .$$

In other words, there is a point $c \in [a, b]$ where f attains its average value for the interval $[a, b]$.



Arclength

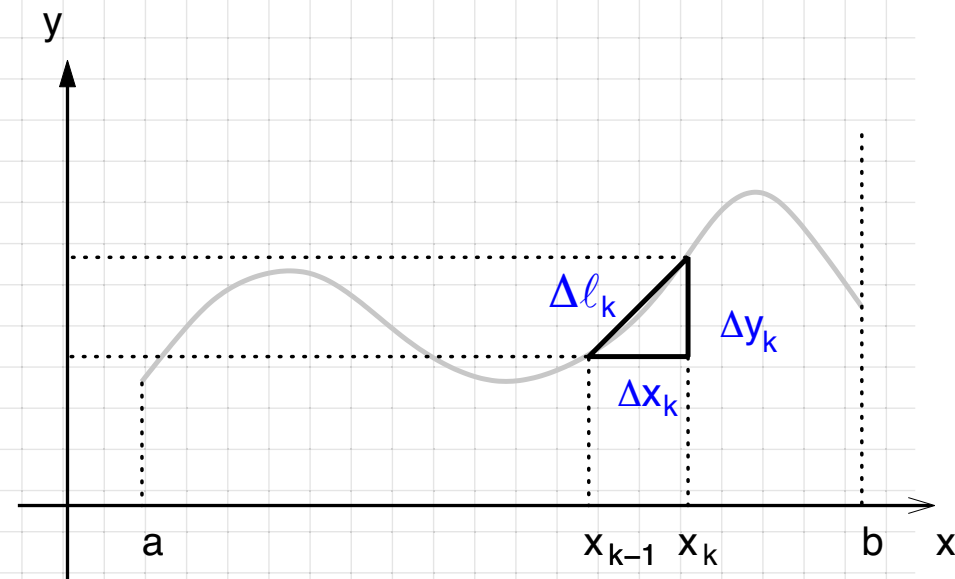
Consider the graph of $y = x^2$. How do we calculate the length of the curve between points $(0, 0)$ and $(1, 1)$?

Slice the curve into small pieces, and approximate the length of each piece by the length $\Delta \ell_k$ of the secant line joining (x_{k-1}, y_{k-1}) to (x_k, y_k) .

$$(\Delta \ell_k)^2 = (\Delta x_k)^2 + (\Delta y_k)^2$$

$$\ell \approx \sum_{k=1}^n \Delta \ell_k = \sum_{k=1}^n \sqrt{(\Delta x_k)^2 + (\Delta y_k)^2} = \sum_{k=1}^n \sqrt{1 + \left(\frac{\Delta y_k}{\Delta x_k}\right)^2} \Delta x_k$$

Consider the limit $n \rightarrow \infty$ then $\ell = \int_a^b d\ell = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$ or, because $y = f(x)$,

$$\ell = \int_a^b \sqrt{1 + [f'(x)]^2} dx$$


Example

Find the length of the curve $y = x^2$ for $x \in [0, 1]$.

Example

Find the length of the curve $y = (e^x + e^{-x})/2$ for $x \in [0, 2]$.

Example

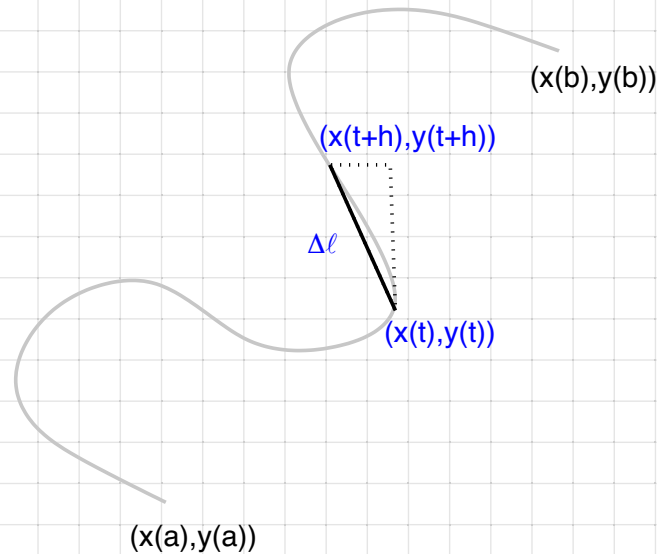
A thin wire of constant linear mass density λ (mass per unit length) lies in the x - y plane along the curve $y = 4\sqrt{2}x^{3/2}/3 - 1$ and $x \in [0, 1]$.

Calculate the mass of the wire.

Parametric Form

The arc length of a curve represented in parametric form $(x(t), y(t))$ between $t = a$ and $t = b$ is given by

$$\ell = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt.$$



$$\Delta \ell = \ell(t+h) - \ell(t) \approx \underbrace{\sqrt{(x(t+h) - x(t))^2 + (y(t+h) - y(t))^2}}_{\text{length of secant line}}$$

$$\ell = \int_a^b d\ell = \int_a^b \sqrt{(dx)^2 + (dy)^2} = \int_a^b \sqrt{(dx/dt)^2 + (dy/dt)^2} dt$$

or in terms of explicit limits:

$$\ell(t+h) - \ell(t) \approx \sqrt{(x(t+h) - x(t))^2 + (y(t+h) - y(t))^2}$$

$$\frac{\ell(t+h) - \ell(t)}{h} \approx \sqrt{\left(\frac{x(t+h) - x(t)}{h}\right)^2 + \left(\frac{y(t+h) - y(t)}{h}\right)^2}$$

Limit $h \rightarrow 0$

$$\begin{aligned} \frac{d\ell}{dt} &= \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} \\ \int_{\ell(a)}^{\ell(b)} d\ell &= \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt \end{aligned}$$

The arc length between $t = a$ and $t = b$ is given by

$$\ell = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt.$$

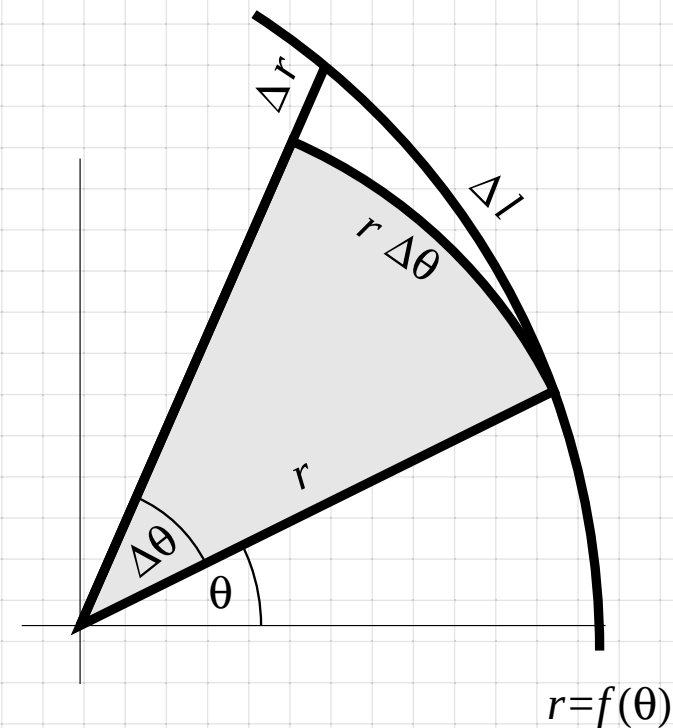
(5.2.4) Polar Curve

The arc length of a polar curve $r = r(\theta)$ is given by

$$\ell = \int_{\theta_1}^{\theta_2} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

$$\begin{aligned} (\Delta\ell)^2 &\simeq (\Delta r)^2 + (r\Delta\theta)^2 \\ &= \left(\left(\frac{\Delta r}{\Delta\theta}\right)^2 + r^2 \right) (\Delta\theta)^2 \end{aligned}$$

Add up all segments $\Delta\ell$



Example

One arch of a *cycloid* is generated by a rolling circle in one revolution around its centre. If the rolling circle sweeps out angle θ , show that the parametric equations for the cycloid are $x = a(\theta - \sin \theta)$ and $y = a(1 - \cos \theta)$, where a is the radius of the circle. Calculate the length of one arch of the cycloid.

Example

Calculate the length of the so-called logarithmic spiral $r = e^{-\theta/10}$ for $\theta \in [0, \infty)$. Is this a finite number?

(5.3) Speed of a Particle along a Planar Curve

Consider a particle moving along a planar curve with position $(x(t), y(t))$ at time t

- ▶ Distance along path or curve (arc length from time 0 to t)

$$s(t) = \int_0^t \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2} du$$

- ▶ Speed

$$v(t) = \frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$$

(Recall the **Fundamental Theorem of Calculus**: $f(t) = \frac{d}{dt} \int_a^t f(u) du$.)

- ▶ Velocity in 2 dimensions (vector in x-y plane)

$$\mathbf{v}(t) \equiv \left(\frac{dx}{dt}, \frac{dy}{dt} \right)$$

Note that $v(t) = |\mathbf{v}(t)|$, the length of the velocity vector.

Example

A particle moves along the path $(x(t), y(t)) = (2t, 1 + t^{3/2})$.

(a) Determine its speed at $t = 1$.

(b) Determine the distance travelled from $t = 0$ to $t = 4$.

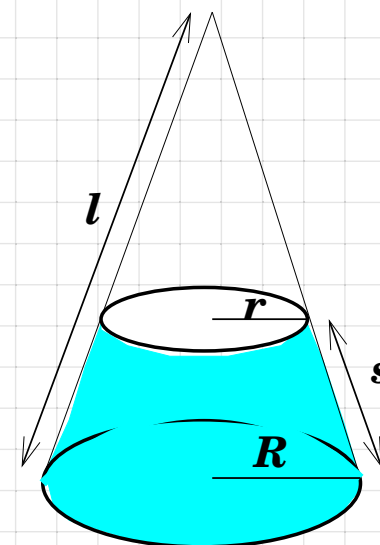
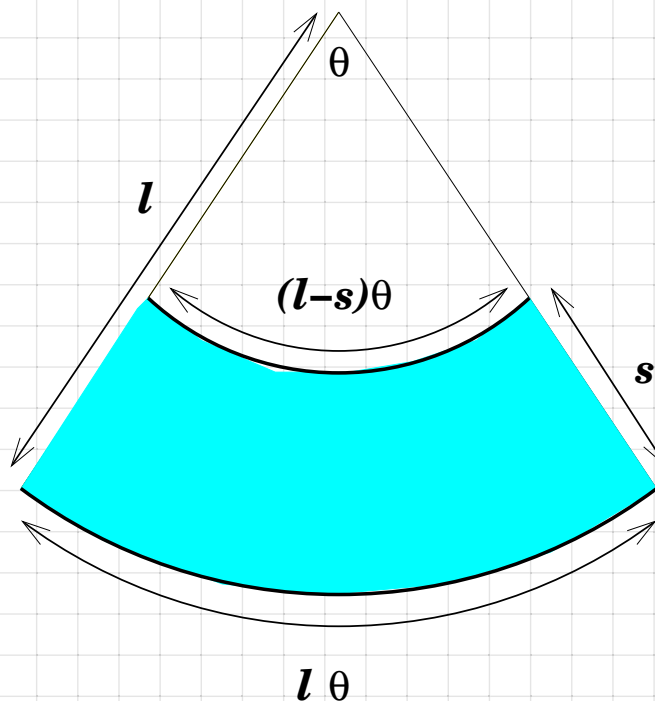
(5.4) Surface Area – Solids of Revolution

Area of frustum $\pi(r + R)s$

$$2\pi r = (\ell - s)\theta$$

$$2\pi R = \ell\theta$$

$$2\pi R - 2\pi r = s\theta$$



The area of the **frustum** is

$$\begin{aligned} A &= \frac{\theta}{2\pi} \pi \ell^2 - \frac{\theta}{2\pi} \pi (\ell - s)^2 = \frac{1}{2\theta} (\ell\theta - (\ell - s)\theta) (\ell\theta + (\ell - s)\theta) \\ &= \frac{1}{2\theta} (2\pi R - 2\pi r) (2\pi R + 2\pi r) = \pi(r + R)s \end{aligned}$$

Surface Area – Sum of Frustum Strip Areas

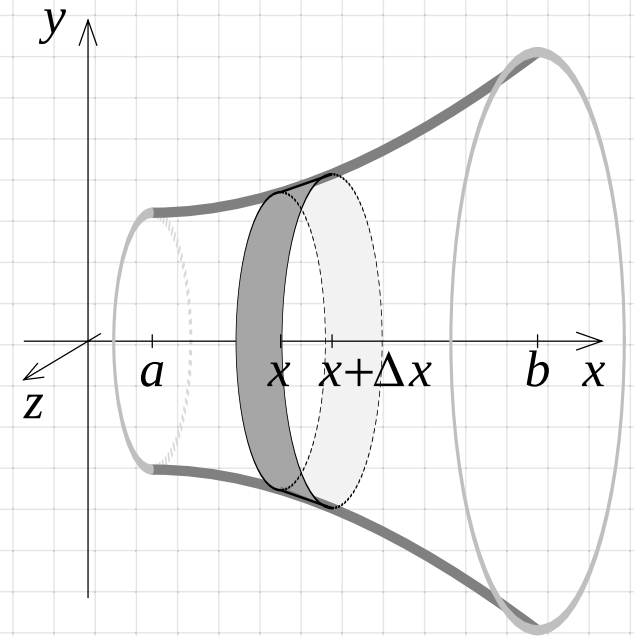
Let $y = f(x)$ be rotated around the x -axis between $x = a$ and $x = b$.

Slice the surface into strips, approximate each strip by a **frustum** with radii y_k and $y_k + \Delta y_k$, and slant length $\Delta \ell = \sqrt{(\Delta x_k)^2 + (\Delta y_k)^2}$.

The approx. area of the k th frustum
 $\pi(y_k + (y_k + \Delta y_k))\sqrt{(\Delta x_k)^2 + (\Delta y_k)^2}$

$$A_k = \pi(2y_k + \Delta y_k) \sqrt{1 + \left(\frac{\Delta y_k}{\Delta x_k}\right)^2} \Delta x_k$$

The **surface area** is the sum of the frustum areas in the limit $\Delta x \rightarrow 0$



Surface Area Formulae

Area of the surface formed by rotating a curve about the x -axis.

Cartesian Form

$$\begin{aligned} S &= \lim_{\Delta x \rightarrow 0} \sum_k \pi(2y + \Delta y) \sqrt{1 + \left(\frac{\Delta y}{\Delta x}\right)^2} \Delta x \\ &= \int_a^b 2\pi y(x) \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \end{aligned}$$

Parametric Form

For parametric curves $(x(t), y(t))$

$$S = \int_a^b 2\pi y(t) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Polar Form

$$x(\theta) = r(\theta) \cos \theta, \quad y(\theta) = r(\theta) \sin \theta$$

$$\begin{aligned} S &= \int_{\theta_1}^{\theta_2} 2\pi y(\theta) \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta \\ &= \int_{\theta_1}^{\theta_2} 2\pi r \sin \theta \sqrt{\left(\frac{dr}{d\theta} \cos \theta - r \sin \theta\right)^2 + \left(\frac{dr}{d\theta} \sin \theta + r \cos \theta\right)^2} d\theta \\ &= \int_{\theta_1}^{\theta_2} 2\pi r \sin \theta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta \end{aligned}$$

Summary: Surface of Area of Revolution about x Axis

- ▶ General

$$\int_a^b 2\pi y \, d\ell$$

- ▶ Cartesian

$$\int_a^b 2\pi y(x) \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

- ▶ Parametric

$$\int_a^b 2\pi y(t) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

- ▶ Polar

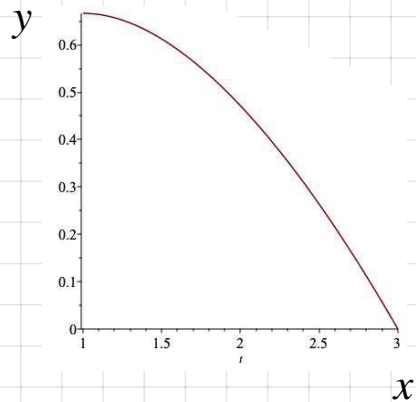
$$\int_{\theta_1}^{\theta_2} 2\pi r \sin \theta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

Example

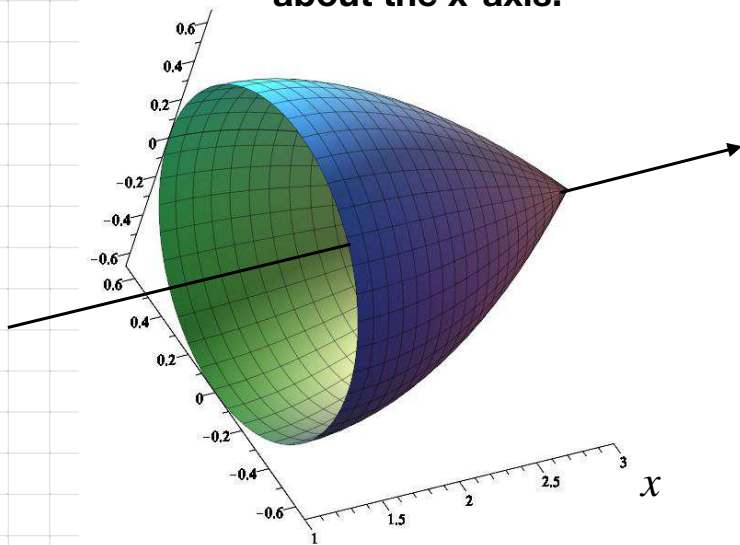
Calculate the surface area of a sphere by rotating a circle of radius R about the x -axis.

Example

Find the area of the surface obtained by rotating $y(x) = x^{1/2} - \frac{1}{3}x^{3/2}$ about the x-axis for $1 \leq x \leq 3$.



**surface of revolution
about the x-axis:**



Volume of revolution about x-axis

$$V = \pi \int_a^b [y(x)]^2 dx$$

Example

Consider “Torricelli’s trumpet”, also known as “Gabriel’s horn” obtained by rotating $f(x) = 1/x$ for $x \in [1, \infty)$ about the x-axis. Determine the area and volume of this solid of revolution.

